Documentation

Arbitrary Waveform Generator

ELT3050 Micro II Final Project

Kevin Kunker, Josh Remillard

Introduction:

An arbitrary waveform generator (AWG), is a specialized piece of electronic test equipment which can synthesize many types of waveforms. For this application, our AWG is capable of generating 3 pre-defined waveforms, as well as 2 custom arbitrary (ARB) waveforms. This documentation will outline key features, necessary hardware, as well as proper operation of the AWG.

Features:

- Output Frequency
  - 1Hz, 10Hz, 100Hz, 1000Hz
- Available Waveforms
  - Sine, Square, Triangle, ARB
- Waveform Attenuation
  - 0 to +3.3 Volts max, Full 12 bit output resolution @ 3.3 volts
- Dual ARB Buffers
  - Current ARB waveform selected will not be overwritten, a second buffer will be used.
- Digital Sync Output
  - Runs in parallel with waveform output, low voltage TTL 0 to +3.3 Volts

Connections:
For this application, a USB connection is suitable. From your computer, connect a USB cable to the Open SDA port on the FRDM-KL25Z as shown.

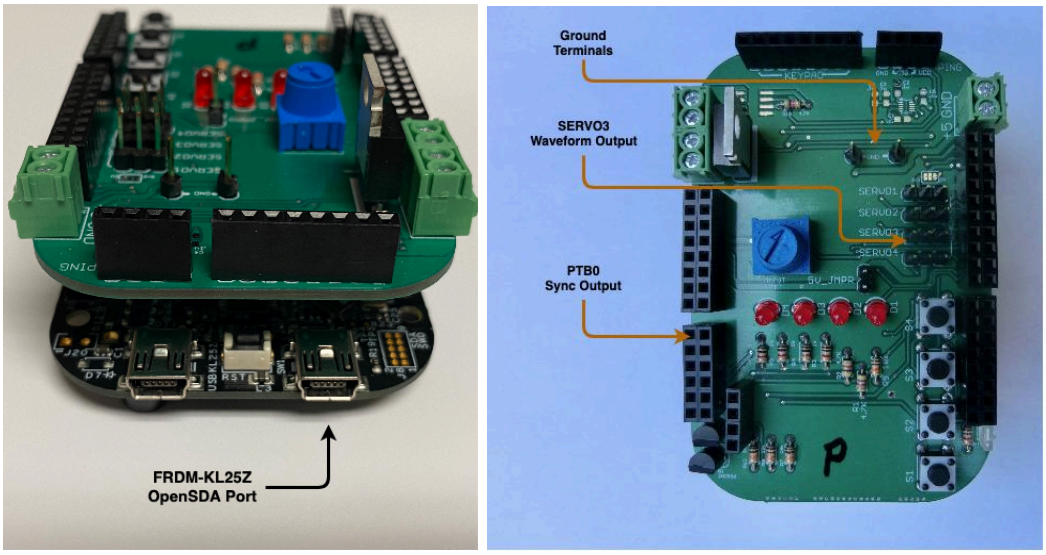

Figure 1. AWG Connection Points

All waveforms will be output on the leftmost pin on the SERVO3 header. In addition, a sync output will be located at port B, pin 0 (PTB0). Ground (GND) connections for both outputs are located directly above the SERVO headers. Either GND terminal can be used. **Note: Be careful not to connect between the +5V and GND pins as this will cause a short-circuit.**

Serial Communication Requirements:
For proper operation, a device capable of serial communication will be necessary. For use with a Windows PC, a terminal emulator such as TeraTerm is suitable. Required serial communication settings will be 9600 baud, 8 bit data, 1 stop bit (as seen in Figure 2).

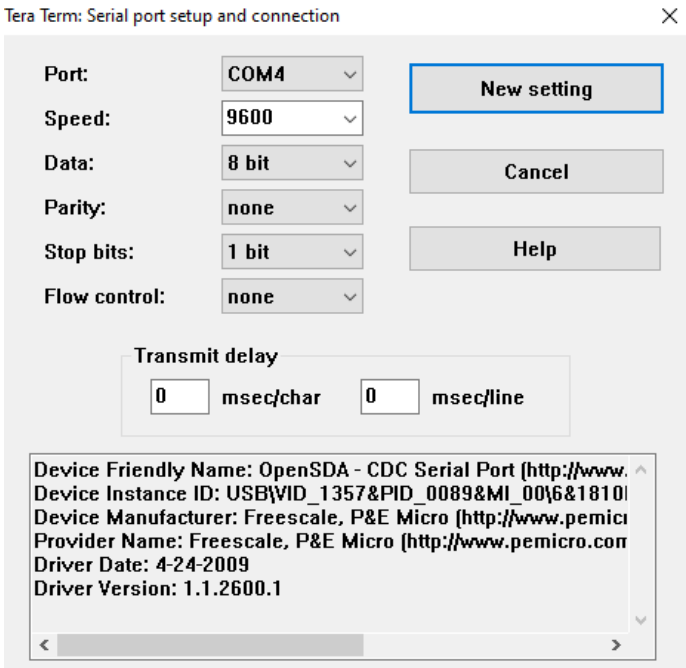

Figure 2. TeraTerm Serial port setup

In TeraTerm, ports are dependent on the computer being used for AWG control. For the example shown, COM4 will be used.
It may also be necessary to configure the terminal settings in order to better facilitate communication with the AWG. In figure 1.1, keystrokes will be echoed on the terminal display. New line settings for both receive, and transmit, have been configured to include both a carriage return (CR), and line feed (LF).

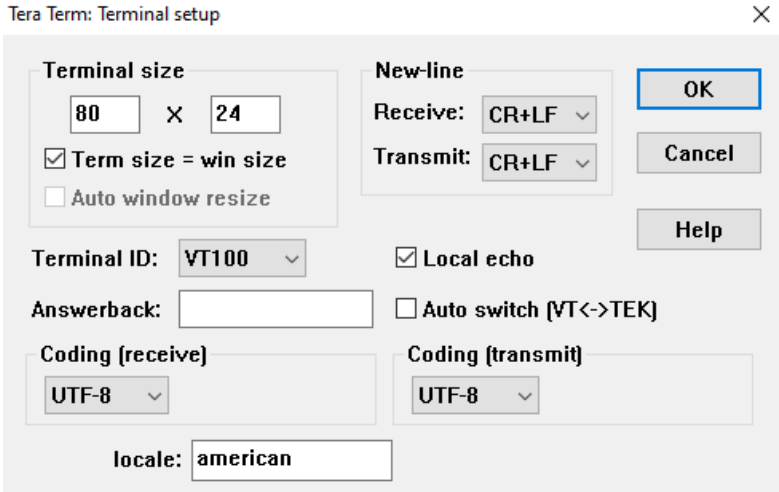

Figure 2.1. Terminal Setup

Commands:
Using the commands listed, operation of the AWG can be changed to meet the requirements of an intended application. All commands must start with "/" followed by the corresponding command ID. Also, be sure to include a space between a command, and any parameters/data. Command ID's and parameters are not case sensitive.

Set Waveform:

Format	/v or /V
Parameters	S,T,Q,A or s,t,q,a
Example	/V T

This command will change the waveform output on the SERVO3 header. Available waveforms include sine (s), triangle (t), square (q), and ARB (a) waveforms. An "/OK" response will confirm successful execution of this command.

Set Frequency:

Format	/f or /F
Parameters	1,10,100,1000
Example	/F 100

This command will change the frequency of the waveform. Available frequencies include 1hz (1), 10Hz (10), 100Hz (100), 1000Hz (1000). An "/OK" response will confirm successful execution of this command.

Set Attenuation:

Format	/a or /A
Parameters	1,2,4,8,16,32,64,128
Example	/A 32

This command will change the attenuation value of the current output waveform. A parameter value of 1 will allow the waveform to run at the full 3.3 volt scale. Each increase in parameter value will decrease the voltage level at which the waveform will be outputted by a factor of two. An "/OK" response will confirm successful execution of this command.

Commands (continued):

Set Current ARB Buffer:

Format	/b or /B
Parameters	A, B, or a, b
Example	/B a

When using ARB waveforms, this command will set either ARB buffer A, or ARB buffer B as the current ARB output buffer. An “/OK” response will confirm successful execution of this command.

ARB Data Write:

Format	/w or /W
Parameters	[Point #] and [Data]
Example	/W 10 3456

When using ARB waveforms, this command will write data to the ARB buffer that is currently not in use. This command requires both a point value and data value. **An ARB buffer contains 100 points ranging from 0 to 99. Each point can hold a data value from 0 to 4095.** The example above will write the data value “3456” to point “10” in the non-current buffer. An “/OK” response will confirm successful execution of this command.

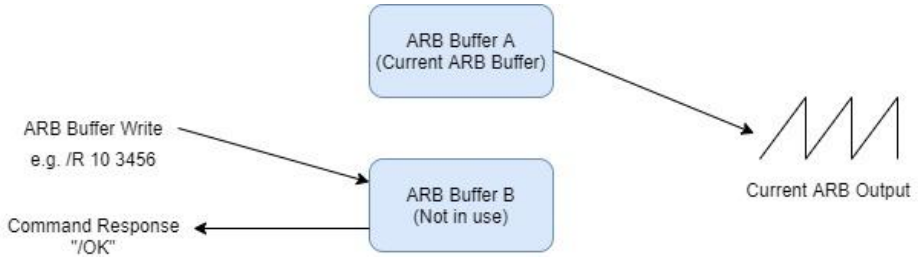

Figure 3.1. ARB Buffer Write Visualization

ARB Data Read:

Format	/r or /R
Parameters	[Point #]
Example	/R 20

When using ARB waveforms, this command will read data from the ARB buffer that is currently not in use. This command requires both a point value and data value. **An ARB buffer contains 100 points ranging from 0 to 99. Each point can hold a data value from 0 to 4095**. The example above will read the data value from point “20” in the non current buffer, and return both the point value and the data value currently at that location.

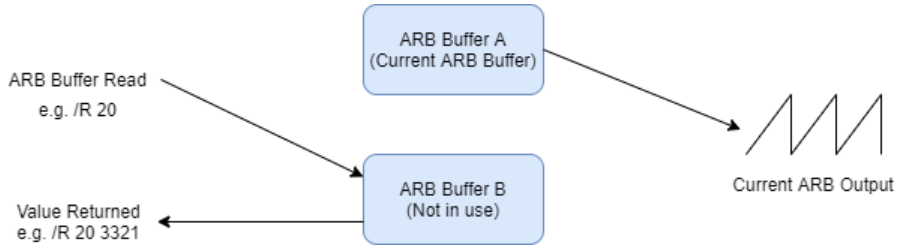

Figure 3.2. ARB Buffer Read Visualization

Commands (continued):

Return Status:

Format	/s or /S
Parameters	none
Example	/S

When performed, this command will allow the user to see the current operating status of the AWG. The returned status will display parameters in the following order:

/S [Frequency] [Attenuation] [Waveform] [Current ARB Buffer]

For example, a return status of "/S 10 2 S B" would indicate that the AWG is currently operating at a frequency of 10Hz. The attenuation is a factor of 2. The current waveform is a sine wave. Buffer B is the current ARB buffer.

Error Codes:

If a command, or parameter does not meet the required attributes, an error code will be sent over the serial port. Refer to the table below when receiving an error code.

Error Code	Description	Error Cause
/E 1	**Unrecognized command**	**Command entered does not match any of the ID's outlined in this documentation**
/E 2	**Unrecognized parameter**	**Parameter entered does not meet the required criteria for specified command ID**
/E 3	**Parameter out of range**	**Parameter value out of range for specified command ID**
/E 4	**Framing error**	**Data frame does not begin with a "/", or contains incorrect spacing between parameters**

Figure 4. Error code table

General Use:

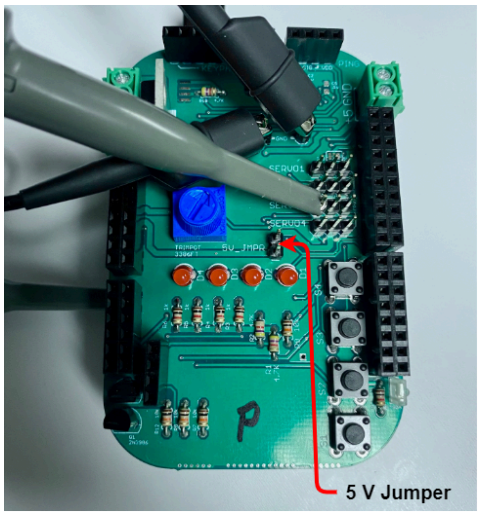

Figure 5. AWG initial setup

Please Note if there is a jumper connected to the 5V terminal, **it will need to be removed**! If the 5V terminal is connected to ground it will cause a short-circuit. Do not make this mistake! After making the necessary connections to the AWG, it is now suitable for waveform generation.

Using sine, square, and triangle waveforms:

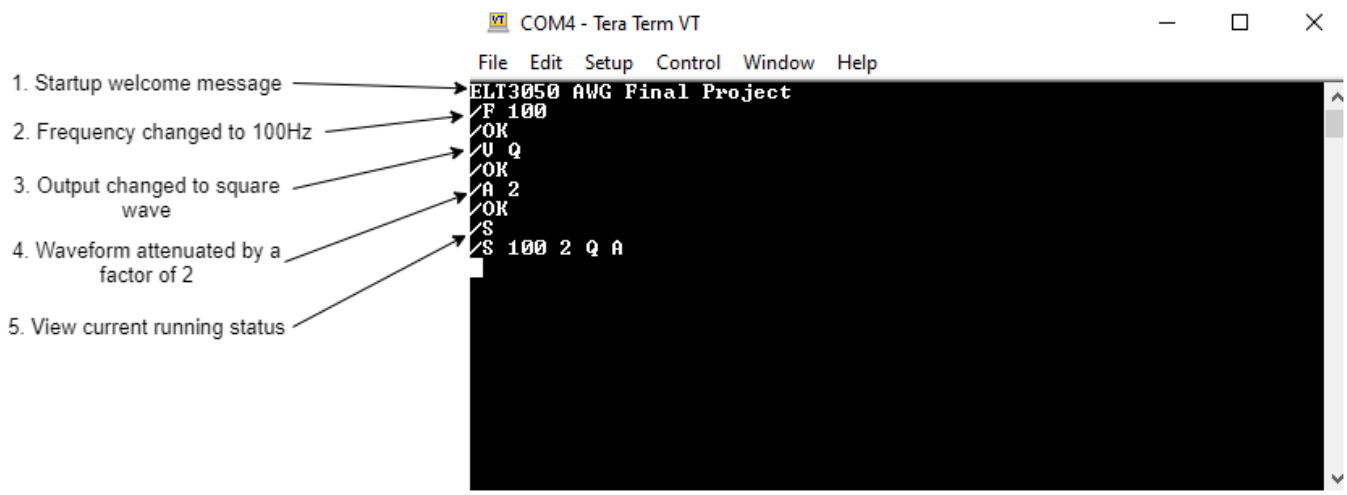

Figure 6, Listed commands in Tera Term

In Figure 6, The Tera Term window depicts a typical scenario when using these types of waveforms. All commands, when entered, must be followed by pressing the enter key in order to properly execute. When using commands, there is no particular order for which they need to operate.

(1) Upon initial startup, the AWG will transmit a welcome message over the serial port. By default, it will output a 1Hz full scale sine wave along with a digital sync signal.
(2) In the scenario above, the command “/F 100” changes the output frequency to 100Hz. A response of “/OK” confirms the command has been properly executed.
(3) The following command “/V Q” will change the output waveform to a square wave having the same characteristics as the previous sine wave. A response of “/OK” confirms the command has been properly executed.

(4) Lastly, the attenuation of the square wave will be modified using “/A 2”. The previous output waveform will be scaled down by a factor of 2. A response of “/OK” confirms the command has been properly executed.

(5) To verify the current running status of the AWG, the command “/S” is entered. The response of “/S 100 2 Q A” verifies that the AWG is sending out a 100Hz square wave. Also, the current ARB buffer is A. This last attribute can be disregarded when working with sine, square, and triangle waveforms.

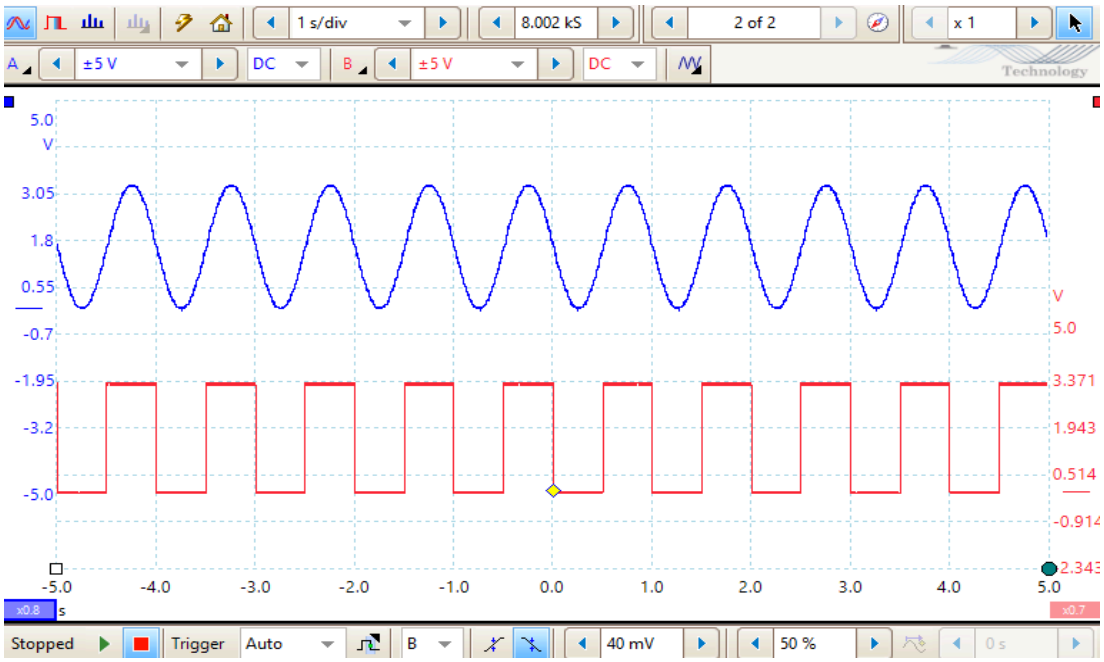

Figure 6.1. Initial AWG waveform after startup
(1 Hz Sine wave at full scale with digital sync)

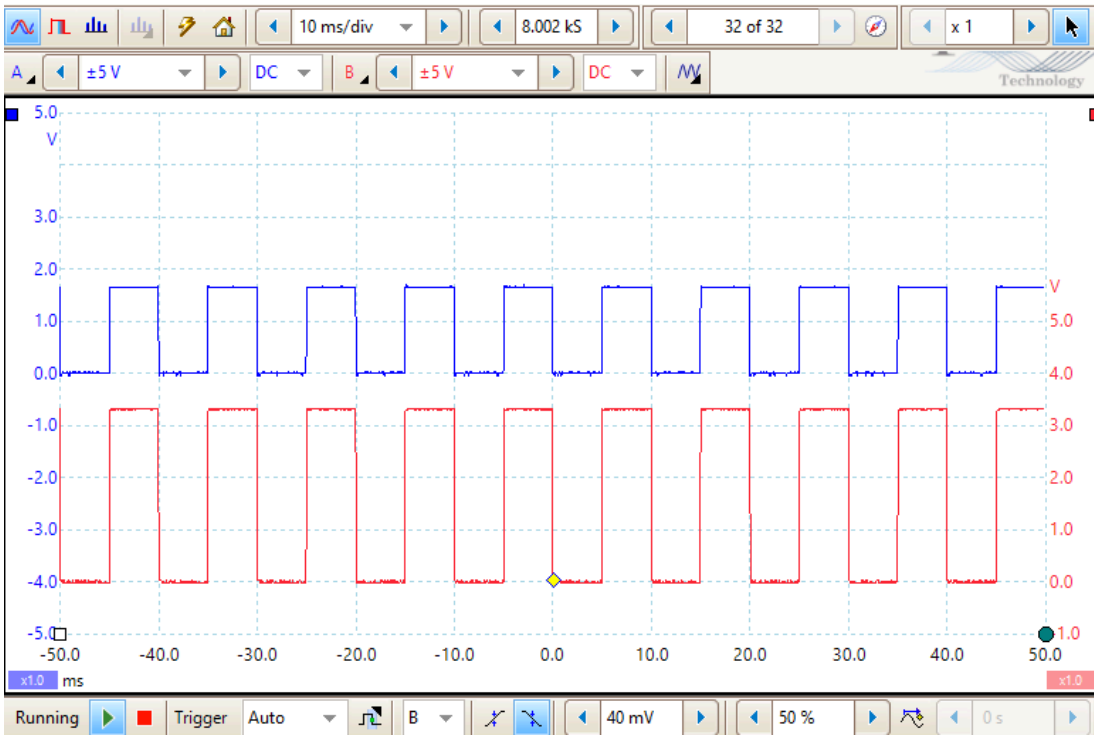

Figure 6.2. Final waveform after executing commands
(100 Hz square wave, attenuated by a factor of 2. ***Note, digital sync always remains at full scale****)*

Using ARB Waveforms:

The AWG has the ability of working with 2 separate arbitrary waveforms, and stores them in dual buffers. While an ARB waveform from one buffer is being sent to the output, the other buffer can be written to with waveform data. Data can also be read from the ARB buffer that is not in use. Data can be entered manually, or via text file containing waveform data. When generating waveform data use the formula.

$$N = \left(\frac{Vout}{3.3V}\right) \times 4096$$

The result, *N*, will be the digital code used when referencing the specified voltage *Vout*. *"N*" can be any value between 0, and 4095.

Note: ARB waveforms are also capable of being attenuated by the same factors as the sine, square, and triangle waveforms. They can also run at the same 1,10,100,1000 Hz frequencies.

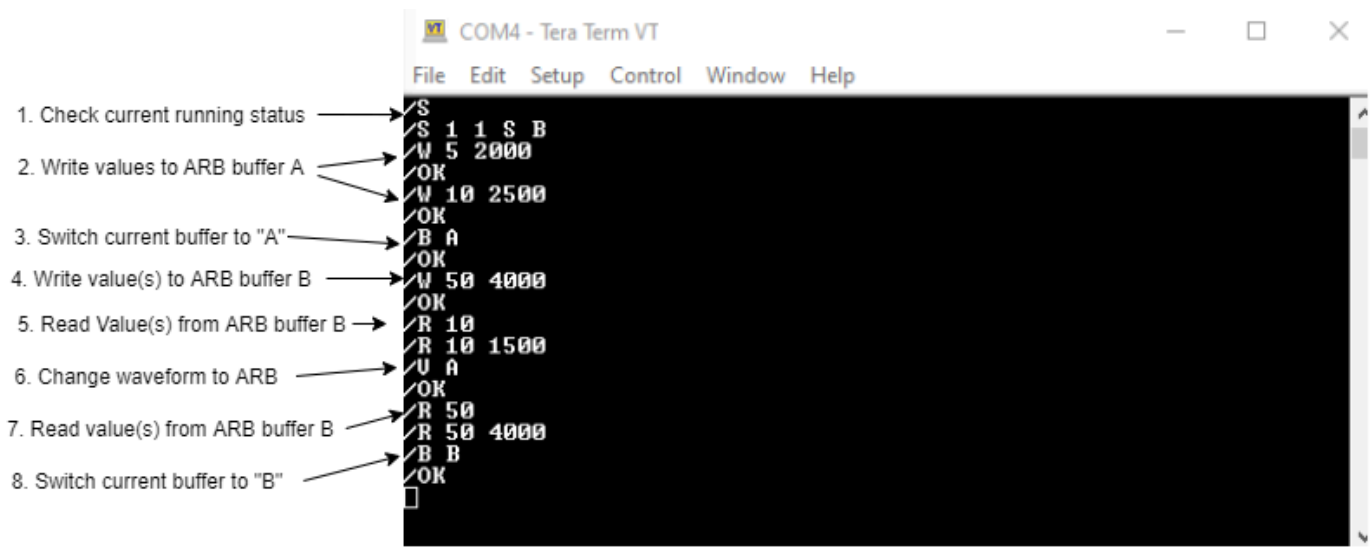

Figure 7. ARB buffer manipulation, ARB data manipulation

Figure 7 shows a typical Tera Term window that has executed commands to manipulate both ARB buffers.

(1) The "/S" command returns the current running status. ARB buffer "B" is the current ARB buffer. Moving forward, data write, and read will have access to ARB buffer "A". Notice that a sine waveform is currently sent to the output; ARB data write, and read can be used anytime in no particular order.
(2) The "/W" command is used to input data into ARB buffer "A". Each ARB buffer contains 100 points where data can be written. For each successful data write, a confirmation message of "/OK" will be sent to the terminal.
(3) Now that some data points have been populated to ARB buffer "A", it will be designated as the current ARB buffer using the "/B A" command. Moving forward, data write, and read will have access to ARB buffer "B". A confirmation message of "/OK" will be sent if the switch is successful.
(4) Since ARB buffer "A" is now the current ARB buffer, The "/W" command will now be used to input data into ARB buffer "B".

Using ARB Waveforms (continued):

(5) In order to verify data entered into ARB buffer “B”, the “/R” command followed by a data point is executed. A return message “/R 10 1500” confirms the data value at point 10 in ARB buffer “B”.

(6) Up until this point, the AWG has been sending a sine waveform to the SERVO3 output. The command “/V A” will switch the waveform output to use the current ARB waveform. In this case, ARB buffer “A”.

(7) While ARB buffer “A” is the current buffer, and is being sent to output. The data stored in ARB buffer “B” can be verified using the “/R” command. The return value of “/R 50 4000” verifies the data located in point 50.

(8) The AWG has the capability to switch ARB buffers on the fly. Up until now, ARB buffer “A” has been sent to the waveform output. The command “/B B” will make ARB buffer “B” the current ARB buffer, and will be sent to the waveform output.

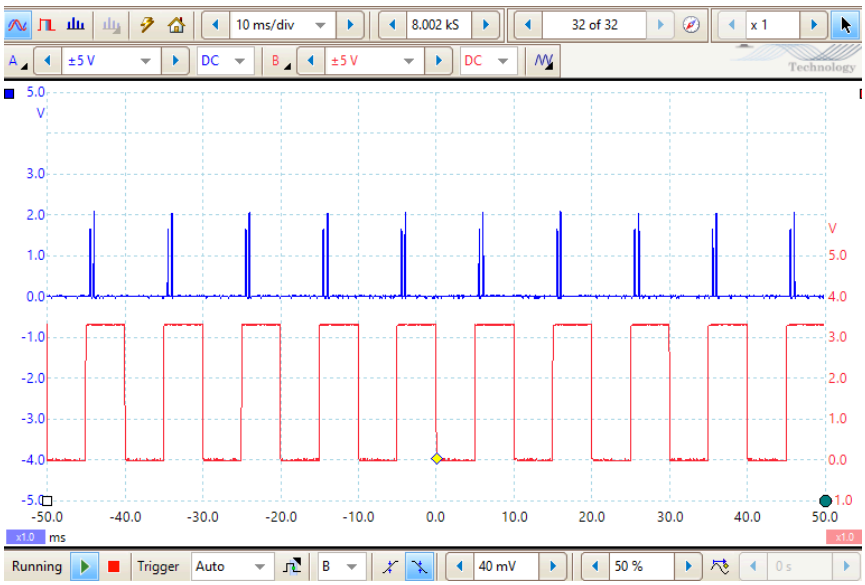

Figure 7.1. ARB buffer “A” full scale waveform output

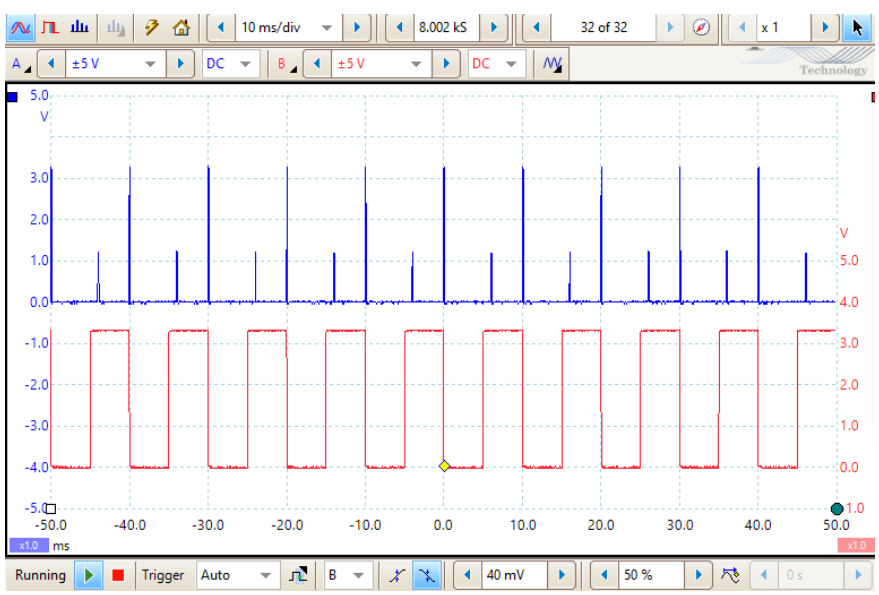

Figure 7.2. ARB buffer “B” full scale waveform output

Using ARB Waveforms (continued):

Sending a text file:

A text file containing write commands with necessary parameters can be sent to the non current ARB buffer in addition to manual data entry. Using Tera Term, navigate to the “File” dropdown menu and select “send file”. For each line in the text file containing a write command, an “/OK” confirmation will be sent.

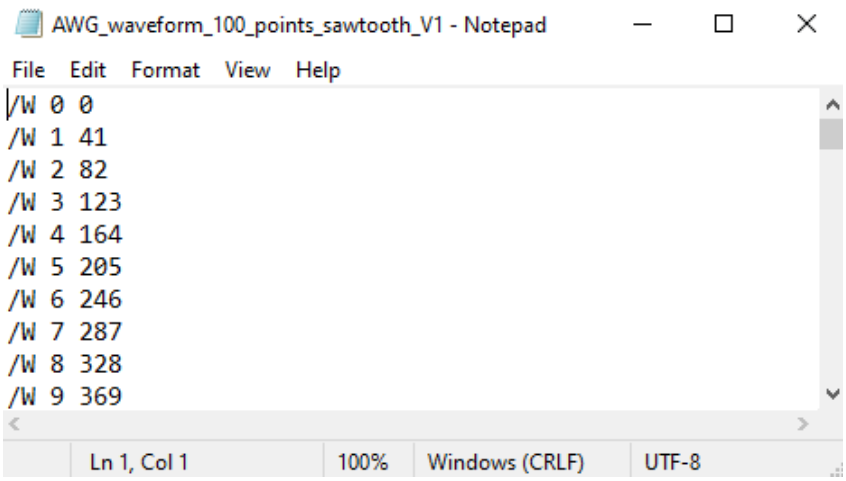

Figure 7.3. ARB write commands in a text file

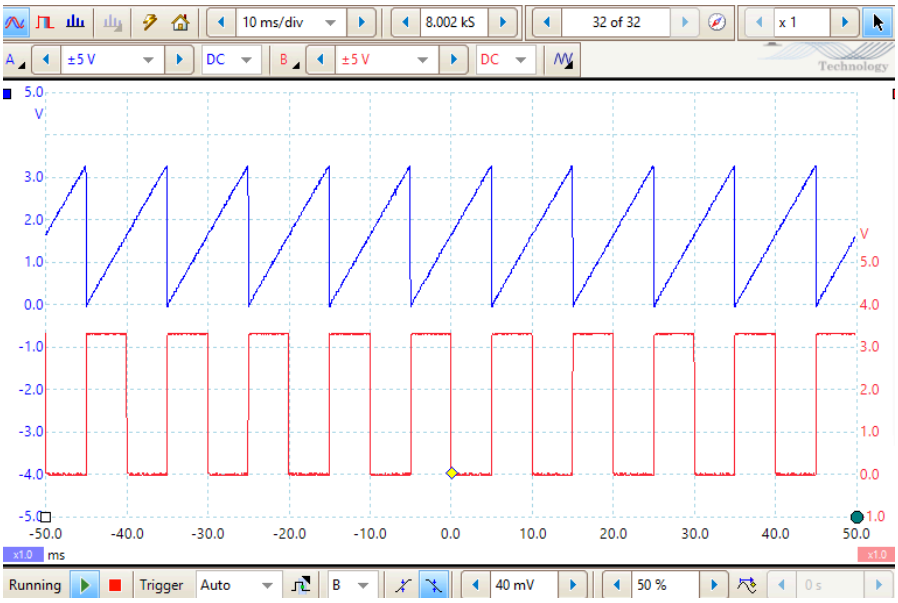

Figure 7.4. ARB Sawtooth full scale waveform from text file upload